Medi-Caps University

Paper Code:EC3CO07

Paper Name:Digital Electronics

From Each Unit: Minimum 120 questions and as many as possible, if it is 240 questions and more it will be well

Sl No	Unit	Question No	Question
1	**I**	1	Which number system has a base 16 Ans- hexadecimal
2	**I**	2	What is a digital-to-analog converter? Ans- a **digital**-to-**analog converter** is a system that converts a **digital** signal into an **analog** signal
3	**I**	3	number (1E.43)16 in octal? $(17103)_8$
4	**I**	4	How many entries will be in the truth table of a 4-input NAND gate Ans- For 4 inputs= 2^4 = 16 truth table entries are necesarry
5	**I**	5	How many bits are needed to store one BCD digit? Ans- 4 Bits
6	**I**	6	Convert (312)8 into decimal Ans- $(202)_{10}$
7	**I**	7	What is the binary subtraction of 101001 - 010110 =? Ans- 010011
8	**I**	8	What is the binary multiplication of 10100 * 01011 =? Ans- 11011100
9	**I**	9	Divide the binary number: 111001 ÷ 1101 and find the reminder Ans- 100
10	**I**	10	1's complement of 1011001 is Ans- 0100110
11	**I**	11	2's complement of 1011011 is Ans-0100101
12	**I**	12	The excess-3 code for 584 is given by Ans – 0111 0100 1000
13	**I**	13	Convert the binary number 11001 to decimal Ans- $(25)_{10}$
14	**I**	14	hexadecimal number (1E.53)16 convert to octal Ans- $(36.246)_8$
15	**I**	15	octal number (651.124)8 to hexadecimal Ans- $(1A9.2A)_{16}$

16	**I**	16	The octal equivalent of the decimal number (417)10 Ans- $(641)_8$

17	**I**	17	Convert the hexadecimal number (1E2)16 to decimal. Ans- 482
18	**I**	18	(170)10 convert to hexadecimal Ans- $(AA)_{16}$
19	**I**	19	Convert (214)8 into decimal. Ans- $(140)_{10}$
20	**I**	20	Convert (0.345)10 into an octal number. Ans- $(0.26050)_8$
21	**I**	21	Convert the binary number (01011.1011)2 into decimal. Ans- $(11.6875)_{10}$
22	**I**	22	Octal to binary conversion: (24)8 =? Ans- 010111
23	**I**	23	Convert binary to octal: (110110001010)2 =? Ans- $(6612)_8$
24	**I**	24	What is the addition of the binary numbers 11011011010 and 010100101? Ans- 11101111111
25	**I**	25	Perform binary addition: 101101 + 011011 = ? Ans- 1001000
26	**I**	26	Perform binary subtraction: 101111 – 010101 = ? Ans- 011010
27	**I**	27	Binary subtraction of 100101 – 011110 is? Ans- 000111
28	**I**	28	Perform multiplication of the binary numbers: 01001 × 01011 = ? Ans- 001100011
29	**I**	29	100101 × 0110 = ? Ans - 101111110
30	**I**	30	On multiplication of (10.10) and (01.01), we get Ans- 011.0010
31	**I**	31	Divide the binary numbers: 111101 ÷ 1001 and find the remainder. Ans- 0111
32	**I**	32	Divide the binary number (011010000) by (0101) and find the quotient. Ans- 101001
33	**I**	33	Binary subtraction of 101101 – 001011 = ? Ans- 100010

34	**I**	34	1’s complement of 1011101 is _____ Ans- 0100010
35	**I**	35	2’s complement of 11001011 is _____ Ans- 00110101
36	**I**	36	On subtracting (01010)2 from (11110)2 using 1’s complement, we get _____ Ans - 10100

37	**I**	37	On subtracting (010110)2 from (1011001)2 using 2’s complement, we get Ans - 1000011
38	**I**	38	On subtracting (001100)2 from (101001)2 using 2’s complement, we get Ans - 011101
39	**I**	39	On addition of 28 and 18 using 2’s complement, we get _____ Ans - 0101110

40	**I**	40	On addition of +38 and -20 using 2's complement, we get _____ Ans - 010010
41	**I**	41	On addition of -46 and +28 using 2's complement, we get _____ Ans- -10010
42	**I**	42	Binary coded decimal is a combination of _____ Ans – 4 binary digits
43	**I**	43	The decimal number 10 is represented in its BCD form as _____ Ans – 0001 0000
44	**I**	44	Add the two BCD numbers: 1001 + 0100 = ? Ans - 00010011
45	**I**	45	Carry out BCD subtraction for (68) – (61) using 10's complement method. Ans - 00000111
46	**I**	46	Code is a symbolic representation of_____information. Ans – discrete
47	**I**	47	When numbers, letters or words are represented by a special group of symbols, this process is called Ans - Encoding
48	**I**	48	A three digit decimal number requires_____for representation in the conventional BCD format Ans – 12 bits
49	**I**	49	How many bits would be required to encode decimal numbers 0 to 9999 in straight binary codes? Ans - 14
50	**I**	50	The excess-3 code for 597 is given by _____ Ans - **100011001010**.
51	**I**	51	The decimal equivalent of the excess-3 number 110010100011.01110101 is Ans – $(970.42)_{10}$
52	**I**	52	According to boolean law: A + 1 = ? Ans - 1
53	**I**	53	The involution of A is equal to _____ Ans - A
54	**I**	54	A(A + B) = ? Ans – A.A + A.B
55	**I**	55	1's complement can be easily obtained by using _____ Ans- Inverter
56	**I**	56	The advantage of 2's complement system is that _____ Ans - Only one arithmetic operation is required

57	**I**	57	The 1's complements requires _____ Ans – One Operation
58	**I**	58	Which logic is used for logical manipulations? Ans - 1's complement

59	**I**	59	The binary number 10101 is equivalent to decimal number …….. Ans- 21
60	**I**	60	2's complement of binary number 0101 is …….. Ans - 1011
61	**I**	61	Decimal number 10 is equal to binary number …… Ans - 1010
62	**I**	62	Which statements does NOT describe an advantage of digital technology? Ans - The values may vary over a continuous range.
63	**I**	63	What is a digital-to-analog converter? Ans - a **digital**-to-**analog converter** is a system that converts a **digital** signal into an **analog** signal
64	**I**	64	What are the symbols used to represent digits in the binary number system? Ans – 0,1
65	**I**	65	Give the decimal value of binary 10010. Ans - 18
66	**I**	66	Convert the fractional binary number 0000.1010 to decimal. Ans – 0.625
67	**I**	67	How we write positive logic Ans - Positive logic is defined as **a high voltage level representing a logic 1 and a low voltage level representing a logic 0**.
68	**I**	68	Convert hexadecimal value 16 to decimal. Ans - 22
69	**I**	69	Convert binary 111111110010 to hexadecimal. Ans – FF216
70	**I**	70	Digital representations of numerical values of quantities may BEST be described as having characteristics: Ans - that vary in discrete steps in proportion to the values they represent.
71	**I**	71	Advantages of parallel transmission of digital data? Ans - High speed − It can transmit N bits at the same time. ... Match to underlying hardware − Internally, computer and transmission hardware use parallel circuits. ... Data is sent much faster as parallel transmission can increase the transfer speed by a factor of n over the serial transmission.
72	**I**	72	Convert the binary number (1011)2 to decimal.

			Ans- 11
73	**I**	73	Are voltages in digital electronics are continuously variable? Ans – YES
74	**I**	74	How expressions is written in the sum-of-products (SOP) form? Ans - he expression which is in the form of SOP is **AB+CD**.
75	**I**	75	Determine the values of A, B, C, and D that make the sum term (A'+B+C'+D) equal to zero. Ans - A = 1, B = 0, C = 1, D = 0
76	**I**	76	How SOP expression to convert an equivalent POS expression?

			Ans- For SOP, we pair 1 and write the equation of pairing in SOP while that can be converted into POS by **pairing 0 in it** and writing the equation in POS form. For example, for SOP if we write x ,y ,z then for pos we write x+y+z.
77	**I**	77	One of De Morgan's theorems states that . Simply stated, this means that logically there is no difference between: Ans - a NOR and an AND gate with inverted inputs
78	**I**	78	Applying DeMorgan's theorem to the expression (A+B)' , we get Ans - _____.
79	**I**	79	The systematic reduction of logic circuits is accomplished by: Ans - The systematic reduction of logic circuits is accomplished by using **boolean algebra**.
80	**I**	80	Is output expression Y=(A+B)(C+D) might indicate a product-of-sums circuit construction? Ans - YES

81	**I**	81	For the SOP expression (AB'C+ABC'+A'BC) , how many 1s are in the truth table's output column? Ans - 3
82	**I**	82	Determine the values of A, B, C, and D that make the product term (A'BC'D) equal to 1. Ans- A=1,B=1,C=1,D=1
83	**I**	83	What is the primary motivation for using Boolean algebra to simplify logic expressions? Ans – A) It may make it easier to understand the overall function of the circuit. B) It may reduce the number of gates. C) It may reduce the number of inputs required.
84	**I**	84	The given expresion AC + ABC = AC is True or False Ans - TRUE
85	**I**	85	When A' and B' are the inputs to a NAND gate, according to De Morgan's theorem, the output expression could be: Ans – X = A+B
86	**I**	86	Which Boolean algebra property allows us to group operands in an expression in any order without affecting the results of the operation [for example, A + B = B + A]? Ans - COMMUTATIVE
87	**I**	87	Why the NAND or NOR gates are referred to as "universal" gates? Ans - The NAND gate and the NOR gate can be said to be universal gates **since combinations of them can be used to accomplish any of the basic operationsand can thus produce an inverter, an OR gate or an AND gate**.
88	**I**	88	The truth table for the three variables SOP expression has make how many input combinations? Ans - 8
89	**I**	89	A Karnaugh map is a systematic way of reducing which type of expression? Ans - The Karnaugh map (KM or K-map) is a method of simplifying **Boolean algebra expressions**.
90	**I**	90	The Boolean expression (A'+B'+C') is logically equivalent to what single gate? Ans - NAND
91	**I**	91	What is the most important operation in binary-coded decimal (BCD) arithmetic? Ans - ADDITION
92	**I**	92	What is a Binary Number System Ans – A method of representing numbers that has 2 as its base and uses only the digits 0 and 1
93	**I**	93	What is Bit in Binary Number? Ans – A bit is the smallest unit of data in a computer
94	**I**	94	Binary equivalent of decimal number 10 Ans - 1010

95	**I**	95	Explain Binary Arithmetic Operations Ans -Binary arithmetic includes the basic arithmetic operations of addition,subtraction,multiplication and division
96	**I**	96	Explain Binary Subtraction
			Ans – Binary subtraction is the process of subtracting binary numbers
97	**I**	97	Subtract (1101)2, and(1010)2 Ans - 0011
98	**I**	98	Divide (1010)2 by (10)2 Ans - 101
99	**I**	99	According to boolean law: A + 1 = ? Ans – 1
100	**I**	100	DeMorgan's theorem states that _____ Ans – the theorem explains that the complement of the product of all the terms is equal to the sum of the complement of each term
101	**I**	101	Complement of the expression A'B + CD' is _____ Ans – (A+B').(C'+D)
102	**I**	102	Simplify Y = AB' + (A' + B)C. Ans – AB' + C
103	**I**	103	Explain what is meant by radix (or base) of a number system? Ans – The radix or base is the number of unique digits, including the digit zero,used to represent numbers .
104	**I**	104	Explain what is meant by LSB and MSB? Ans - In a binary number, the bit furthest to the left is called the most significant bit(MSB) and the bit furthest to the right is called is called the least significant bit (LSB)
105	**I**	105	Explain what are octal number systems? Ans – A number system which has its base as 8 is called an octal number system . It uses numbers from 0 to 7
106	**I**	106	Explain what are hexadecimal systems? Ans – The hexadecimal number system is a type of number system, that has a base equal to 16.
107	**I**	107	Let x be a Boolean variable and x' its complement. Then x + x′ = Ans - 1
108	**I**	108	Let x be a Boolean variable and x' its complement. Then x x′ = Ans – 0
109	**I**	109	Let x be a Boolean variable and x' its complement. Then 1 + x = Ans – 1

110	**I**	110	Convert decimal 23 into its binary equivalent number Ans - 10111
111	**I**	111	Determine the binary equivalent of decimal number 17. Ans - 10001
112	**I**	112	Convert decimal 0.8125 into its equivalent binary number Ans – 0.1101
113	**I**	113	Convert 0.65 into its binary fraction Ans – 0.10100110011
114	**I**	114	Convert binary 11001 into octal equivalent Ans - 31
115	**I**	115	Convert binary 0.111 001110 into its equivalent octal Ans – $(0.716)_8$
116	**I**	116	Convert 1111 1110 0011 into hexadecimal equivalent Ans – FE3
117	**I**	117	Convert 1111 10001.100 1100 1101 into hexadecimal equivalent Ans – 497.60009765625
118	**I**	118	Convert Hex number 6D5 into its binary equivalent. Ans - 11011010101
119	**I**	119	Explain Application of Binary Numbers in Computers Ans- The binary number system, also called the base-2 number system, is a method of representing numbers that counts by using combinations of only two numerals: zero(0) and one(1). Computers use the binary number system to manipulate and store all of their data including numbers, words, videos , graphics and music
120	**I**	120	one's complement 1000 will be Ans - 0111

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1	**II**	1	What are universal gates? Ans :- A universal gate is a gate which can implement any Boolean function without need to use any other gate type. The NAND and NOR gates are universal gates.
2	**II**	2	State the De Morgan's Theorem? Ans:- **De Morgan's First Theorem:-** Statement - The complement of a logical sum equals the logical product of the complements. Logic equation - $A+B^{\overline{}}=A^{\overline{}}.B^{\overline{}}$ **De Morgan's Second Theorem:-** Statement - The complement of a logical product equals the logical sum of the complements. Logic equation - $A.B^{\overline{}}=A^{\overline{}}+B^{\overline{}}$

3	**II**	3	What is a Logic gate? Ans:- logic gate is a device that acts as a building block for digital circuits. They perform basic logical functions that are fundamental to digital circuits.
4	**II**	4	What is meant by K-Map or Karnaugh Map? Ans:- A Karnaugh map (K-map) is a pictorial method used to minimize Boolean expressions without having to use Boolean algebra theorems and equation manipulations. A K-map can be thought of as a special version of a truth table . Using a K-map, expressions with two to four variables are easily minimized.
5	**II**	5	Name the two forms of Boolean expression? Ans:- Sum-of-Products (SOP) Form. Product-of-sums (POS) form.
6	**II**	6	What are Minterm and Maxterm? Ans:- minterm for **each combination of the variables that produces a 1 in the function and then taking the OR of all those terms**. maxterm for each combination of the variables that produces a 0 in the function and then taking the AND of all those terms.
7	**II**	7	Define Pair, Quad, and Octet? Ans:- Quad: A group of 4 one's that are horizontally or vertically adjacent. End to end or in form of a square. **A quad eliminates two variables and their complements**. ... **An octet** eliminates three variables and their complements. **Pair** • A pair can be formed by grouping two horizontal or two vertical '1'. • A pair of '1' reduces 1 variable.
8	**II**	8	What are combinational circuits? Ans:- Combinational circuit is a circuit in which we combine the different gates in the circuit, for example encoder, decoder, multiplexer and demultiplexer.
9	**II**	9	What is Half-Adder? Ans:- The addition of 2 bits is done using a combination circuit called a Half adder.
10	**II**	10	What is Full-Adder? Ans:- Full Adder is the adder which adds three inputs and produces two outputs. The first two inputs are A and B and the third input is an input carry as C-IN. The output carry is designated as C-OUT and the normal output is designated as S which is SUM.
11	**II**	11	What are the applications of Multiplexer (MUX)? Ans:- - In multiplexer, the usage of a number of wires can be decreased - It reduces the cost as well as the complexity of the circuit
12	**II**	12	What is a Demultiplexer? Ans:- A demultiplexer (also known as a demux or data distributor) is defined as a circuit that can distribute or deliver multiple outputs from a single input
13	**II**	13	Define edge triggered flip flop? Ans:- An edge triggered flip-flop (or just flip-flop in this text) is a modification to the latch which allows the state to only change during a small period of time when the clock pulse is changing from 0 to 1. It is said to trigger on the edge of the clock pulse, and thus is called an edge-triggered flip-flop

14	**II**	14	Define Half Subtractor Ans:- The half subtractor is a combinational circuit which is used to perform subtraction of two bits.
15	**II**	15	Define full Subtractor Ans:- The full subtractor is a combinational circuit which is used to perform subtraction of three input bits: the minuend , subtrahend , and borrow in . The full

			subtractor generates two output bits: the difference and borrow out .
16	**II**	16	what is boolean laws Ans:- The basic **Laws of Boolean Algebra** that relate to the *Commutative Law* allowing a change in position for addition and multiplication, the *Associative Law* allowing the removal of brackets for addition and multiplication, as well as the *Distributive Law* allowing the factoring of an expression, are the same as in ordinary algebra. Each of the *Boolean Laws* above are given with just a single or two variables, but the number of variables defined by a single law is not limited to this as there can be an infinite number of variables as inputs too the expression.
17	**II**	17	The output of a NOR gate is HIGH if_____ Ans:-. if both the inputs to the gate are LOW
18	**II**	18	why exclusive gates are exclusive Ans:- This gate is called XOR or exclusive OR gate because **its output is only 1 when its input is exclusively 1**.
19	**II**	19	What is encoder? Ans:- Device which converts an input device state into a binary representation of ones or zeros
20	**II**	20	What is decoder? Ans:- decoder is a combinational logic circuit that converts binary information from the n coded inputs to a maximum of 2^n unique outputs.
21	**II**	21	The NOR gate is OR gate followed by …… Ans:- NOT gate
22	**II**	22	In Boolean algebra, the bar sign (-) indicates …….. Ans:- NOT Operation
23	**II**	23	Which of the gate will returns a 1 only if both the inputs are 0? Ans:- NAND Gate
24	**II**	24	If A and B are the inputs of a half adder, the sum is given by _ Ans:- **A XOR B**,
25	**II**	25	If A and B are the inputs of a half adder, the carry is given by _____ Ans:- A AND B
26	**II**	26	In which operation carry is obtained? Ans:- IN Addition
27	**II**	27	The difference between half adder and full adder is _____ Ans:- Half adder has two inputs while full adder has **three outputs**;
28	**II**	28	How many AND, OR and Ex-OR gates are required for the configuration of full adder? Ans:- There are **2 AND, 1 OR and 2 EXOR gates** required for the configuration of full adder,
29	**II**	29	Half-adders have a major limitation in that they cannot _____ Ans:- they **cannot accept a carry bit from a previous stage**, meaning that they cannot be chained together to add multi-bit numbers.

30	**II**	30	If A, B and C are the inputs of a full adder then the sum is given by ___ Ans:- **A XOR B XOR C**.
31	**II**	31	Which one is not the possible outcome of magnitude comparator is ___ Ans:- A comparator is a combinational circuit that takes two numbers as input in binary form and results whether one input is greater, lesser or equal to the other input. In a digital comparator, only 3 outputs are possible (i.e. A = B, A > B, A < B). So, **a – b** is an incorrect option
32	**II**	32	Which one is a basic comparator? Ans:- **an XNOR gate** is a basic comparator, because its output is "1" only if its two input bits are equal.
33	**II**	33	The purpose of a Digital Comparator is_____. Ans:- The purpose of a Digital Comparator is **to compare a set of variables or unknown numbers**, for example A (A1, A2, A3, …. An, etc) against that of a constant
			or unknown value such as B (B1, B2, B3, …. Bn, etc) and produce an output condition or flag depending upon the result of the comparison.
34	**II**	34	A magnitude comparator is defined as a digital comparator which has__. Ans:- A Magnitude Comparator is a digital comparator which has **three output terminals**, one each for equality A = B, greater than A > B and less than A < B.
35	**II**	35	In a comparator, if we get input as A>B then the output will be_____. Ans:- **output will be 1**.as it satisfies one of the condition among three.
36	**II**	36	How many inputs are required for a digital comparator ? Ans:- 2
37	**II**	37	An identify comparator is defined as a digital comparator which has___. Ans:- **only one output terminal for when A = B either "HIGH" A = B = 1 or "LOW" A = B = 0**.
38	**II**	38	Gray to binary conversion can be implemented with _____ . Ans:-
39	**II**	39	A binary code that progresses such that only one bit changes between two successive codes is : Ans:- Reflected Binary Code or Gray Code
40	**II**	40	Convert binary number into gray code: 100101 Ans:- 110111
41	**II**	41	Reflected binary code is also known as_____. Ans:- **gray code** because one digit reflected to the next bit.
42	**II**	42	Convert gray code into binary code: 0101 . Ans:- 0110
43	**II**	43	A demultiplexer accepts ..... inputs. Ans:- **accepts only one input**
44	**II**	44	In a multiplexer, the selection of a particular input line is controlled by _____ Ans:- **a set of selected lines** in a multiplexer, which helps to select a particular input from several sources
45	**II**	45	Which of the following circuit can be used as parallel to serial converter? Ans:- Multiplexer

46	**II**	46	How many select lines would be required for an 8-line-to-1-line multiplexer? Ans:- 8 input lines mean 2^3 inputs. So, 3 control lines are possible.
47	**II**	47	How many NOT gates are required for the construction of a 4-to-1 multiplexer? Ans:- 2
48	**II**	48	In 1-to-4 demultiplexer, how many select lines are required? Ans:-2
49	**II**	49	How many select lines are required for a 1-to-8 demultiplexer? Ans:-3

50	**II**	50	The enable input is also known as _____ Ans:- **strobe** which is used to cascade two or more multiplexer ICs to construct a multiplexer with a larger number of inputs.
51	**II**	51	BCD to seven segment conversion is a _____ process Ans:- Decoder
52	**II**	52	Invalid BCD can be made to valid BCD by adding with _____ Ans:- **adding 6** to our sum.
53	**II**	53	Device which converts an input device state into a binary representation of ones or zeros is termed as Ans:-Encoder
54	**II**	54	A circuit that changes a code into a set of signals is called Ans:- Decoder
55	**II**	55	A decoder converts n inputs to____outputs Ans:- 2n
56	**II**	56	The inputs of a NAND gate are connected together. The resulting circuit is ……. Ans:- NOT Gate
57	**II**	57	Which equation would accurately describe a four-input OR gate when A = 1, B = 1, C = 0, and D = 0? Ans:-1100=1
58	**II**	58	For subtracting 1 from 0, we use to take a__from neighbouring bits. Ans:- Borrow
59	**II**	59	Let the input of a subtractor is A and B then what the output will be if A = B? Ans:- A = B = 0 or A = B = 1.
60	**II**	60	Let A and B is the input of a subtractor then the difference output will be Ans:- AB’ + BA’.
61	**II**	61	What does AND gate mean? Ans:- Boolean operator which gives the value one if and only if all the operands are one, and otherwise has a value of zero.
62	**II**	62	WHAT IS AND gate example? Ans:- **A & B = B & A**.
63	**II**	63	What is the formula of AND gate? Ans:- **C=A&B**.
64	**II**	64	How does an AND gate work? Ans:- An AND gate is a logic gate having two or more inputs and a single output. An AND gate **operates on logical multiplication rules**. In this gate, if either of the inputs is low (0), then the output is also low. ... An AND gate can have any number of inputs, although 2 input and 3 input AND gates are the most common.
65	**II**	65	Is a gate a door? Ans:-NO
66	**II**	66	What is EX OR gate? Ans:- **digital logic gate that gives a true (1 or HIGH) output when the number of true inputs is** odd
67	**II**	67	What are the basic gates? Ans:- All digital systems can be constructed by only three basic logic gates. These basic gates are called **the AND gate, the OR gate, and the NOT gate**
68	**II**	68	Can AND gate have 3 inputs? Ans:-yes
69	**II**	69	WHAT IS AND OR invert gate? Ans:- AOI gates are two-level compound (or complex) logic functions constructed from the combination of one or more AND gates followed by a NOR gate
70	**II**	70	What are universal gates?

<table>
<tr><td></td><td></td><td></td><td>Ans:- A universal gate is a gate which can implement any Boolean function without need to use any other gate type. The NAND and NOR gates are universal gates.</td></tr>
<tr><td>71</td><td>II</td><td>71</td><td>How many transistors are in AND gate?
Ans:- A two input AND gate has a minimum of 6 transistors.</td></tr>
<tr><td>72</td><td>II</td><td>72</td><td>Why do we use an AND gate?
Ans:- AND gate used to performs a binary multiplication of binary digits</td></tr>
<tr><td>73</td><td>II</td><td>73</td><td>Why are they called logic gates?
Ans:- They are the basic building blocks of any digital system. Hence logic gates are named as AND gate, OR gate, NOT gate, etc.</td></tr>
<tr><td>74</td><td>II</td><td>74</td><td>Which IC is used for OR gate?
Ans:-</td></tr>
<tr><td>75</td><td>II</td><td>75</td><td>What is OR gate truth table?
Ans:- A logical OR operation has a high output (1) if one or both the inputs to the gate are high (1). If neither input is high, a low output (0) results.
<table>
<tr><td>A</td><td>B</td><td>Q</td></tr>
<tr><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td></tr>
</table></td></tr>
<tr><td>76</td><td>II</td><td>76</td><td>Which gate is called an inverter? Ans:-
NOT Gate</td></tr>
<tr><td>77</td><td>II</td><td>77</td><td>What type of IC is 7400?
Ans:- NAND Gates</td></tr>
<tr><td>78</td><td>II</td><td>78</td><td>How is XOR gate different from OR gate?
Ans:- In XOR gate if input is same then output will be low whereas in OX gate if one of the input is high then output will be high.</td></tr>
<tr><td>79</td><td>II</td><td>79</td><td>Why XOR gate is called exclusive?
Ans:- because its boolean expression excludes the “OR BOTH” case of Q = “1” when both A and B = “1”</td></tr>
<tr><td>80</td><td>II</td><td>80</td><td>What is the expression of an EXOR gate?
Ans:- The Boolean expression is written as Q = A XOR B.</td></tr>
<tr><td>81</td><td>II</td><td>81</td><td>What is an adder in digital electronics?
Ans:- An adder is a digital circuit that performs addition of numbers</td></tr>
<tr><td>82</td><td>II</td><td>82</td><td>What is adder and its types?
Ans:- An adder is a device that will add together two bits and give the result as the output. ... There are two kinds of adders - half adders and full adders.
A half adder just adds two bits together and gives a two-bit output. A full adder adds two inputs and a carried input from another adder, and also gives a two-bit output.</td></tr>
</table>

83	**II**	83	How does a adder work? Ans:- A full adder adds three one-bit binary numbers, two operands and a carry bit. The adder outputs two numbers, a sum and a carry bit. The term is contrasted with a half adder, which adds two binary digits.
84	**II**	84	Why full adder is used? Ans:- because it **adds together two binary digits, plus a carry-in digit to produce a sum and carry-out digit**.
85	**II**	85	Which gate is called an NOT ? Ans:- **an inverter or NOT** gate is a logic gate which implements logical negation.

86	**II**	86	What is adder explain it? Ans:- An adder is a device that will add together two bits and give the result as the output. ... There are two kinds of **adders - half adders and full adders**. A half adder just adds two bits together and gives a two-bit output. A full adder adds two inputs and a carried input from another adder, and also gives a two-bit output.
87	**II**	87	What are the applications of adders? Ans:- A Full Adder's circuit can be used as a part of many other larger circuits like Ripple Carry Adder, which adds n-bits simultaneously. The dedicated multiplication circuit uses Full Adder's circuit to perform Carryout Multiplication. Full Adders are used in ALU- Arithmetic Logic Unit.
88	**II**	88	What is 4bit adder? Ans:- ull adder that **performs the addition of two 4-bit binary words**.
89	**II**	89	What is half adder? Ans:- The addition of 2 bits is done using a combination circuit called a Half adder

90	**II**	90	What is full adder? Ans:- Full Adder is the adder which adds three inputs and produces two outputs. The first two inputs are A and B and the third input is an input carry as C-IN. The output carry is designated as C-OUT and the normal output is designated as S which is SUM
91	**II**	91	What is a subtractor in digital electronics? Ans:- **An electronic logic circuit for calculating the difference between two binary numbers, the minuend and the number to be subtracted, the subtrahend**
92	**II**	92	What are the types of subtractors Ans:- **There are two types of subtractors.** Half Subtractor. Full Subtractor.
93	**II**	93	What is difference between multiplexer and demultiplexer? Ans:- A multiplexer is a combinational circuit that provides single output but accepts multiple data inputs. A demultiplexer is a combinational circuit that takes single input but that input can be directed through multiple outputs. It performs parallel to serial conversion.
94	**II**	94	What is multiplexer ? Ans:- A multiplexer is a combinational circuit that provides single output but accepts multiple data inputs.
95	**II**	95	What is demultiplexer? Ans:- A demultiplexer is a combinational circuit that takes single input but that input can be directed through multiple outputs.
96	**II**	96	What is the principle of multiplexer? Ans:- **combining multiple inputs into a single data stream**.
97	**II**	97	Why do we use demultiplexer? Ans:- Demultiplexer is **used to connect a single source to multiple destinations**.
98	**II**	98	Why do we use multiplexer ? Ans:- A multiplexer makes **it possible for several input signals to share one device or resource.**
99	**II**	99	What is a 2 1 multiplexer? Ans:- In 2×1 multiplexer, there are only **two inputs**, i.e., A_0 and A_1, 1 selection line, i.e., S_0 and single outputs, i.e., Y. On the basis of the combination of inputs which are present at the selection line S^0, one of these 2 inputs will be connected to the output.
100	**II**	100	Is multiplexer a decoder or encoder? Ans:- Encoder
101	**II**	101	Why a multiplexer is called a data selector? Ans:- multiplexers are also known as data selectors **because they can "select" each input line, are constructed from individual Analogue Switches encased in a single IC package as** opposed to the "mechanical" type selectors such as normal conventional switches and relays.
102	**II**	102	What is an decoder in digital electronics? Ans:- A decoder is **a combinational circuit that converts binary information from n input lines to a maximum of 2^n unique output lines**.
103	**II**	103	What is a 3 to 8 decoder? Ans:- A 3 to 8 decoder has **three inputs (A,B,C) and eight outputs (DO to D7)**. Based on the 3 inputs one of the eight outputs is selected.
104	**II**	104	What are the types of decoders? Ans:- 2 to 4 line decoder 3 to 8 line decoder 4 to 16 line Decoder
105	**II**	105	Why is decoder used? Ans:- The decoder is an **electronic device that is used to convert a digital signal to an analog signal**. It allows a single input line and produces multiple output lines.

106	**II**	106	How does decoder work? Ans:- A decoder circuit takes binary data of 'n' inputs into '2^n' unique output. In addition to input pins, the decoder has a enable pin
107	**II**	107	What are the characteristics of the decoder? Ans:- decoder is **a combinational logic circuit that takes a binary input, usually in a coded form, and produces a one-bit output, on each of a number of output lines**.
			The logic state (1 or 0) on any of the output lines depends on a particular code appearing on the input lines.
108	**II**	108	What is an encoder in digital electronics? Ans:- Device which converts an input device state into a binary representation of ones or zeros.
109	**II**	109	What is meant by encoder? Ans:- Device which converts an input device state into a binary representation of ones or zeros.
110	**II**	110	What is encoder function? Ans:- Encoders are **used to translate the decimal values to the binary in order to perform the binary functions** such as addition, subtraction, multiplication, etc.
111	**II**	111	What is encoder example? Ans:- For example, in 4-2 encoders, if we give 4 inputs it produces only 2 outputs.
112	**II**	112	What is the advantage of using an encoder? Ans:- **Low-cost feedback. Integrated electronics. Compact in size. Fuses optical and digital technology**.
113	**II**	113	What is a code converter in digital electronics? Ans:- A code converter circuit **will convert coded information in one form to a different coding form**. One example of code conversion is to convert BCD to straight binary.
114	**II**	114	Give one example of code convertor. Ans:- One example of code conversion is to convert BCD to straight binary.
115	**II**	115	What does code conversion mean? Ans:- The Code converter is **used to convert one type of binary code to another**
116	**II**	116	list the types of code converter? Ans:- Binary to BCD code converter. BCD to Excess-3 code converter. BCD to Gray code converter. Gray code to Excess-3 code converter.
117	**II**	117	What is magnitude comparator in digital electronics? Ans:- magnitude comparator is **a hardware electronic device that takes two numbers as input in binary form and determines whether one number is greater than, less than or equal to the other number**.
118	**II**	118	Where do you use magnitude comparator Ans:-
119	**II**	119	How many types of comparator are there? Ans There are two main types of digital comparators available: **Identity comparator and Magnitude comparator**.
120	**II**	120	What is 2 bit digital comparator? Ans:- A 2-bit comparator **compares two binary numbers, each of two bits and produces their relation such** as one number is equal or greater than or less than the other.

1	**III**	1	What are sequential circuits? Output depends not only on current input but also on past input values, e.g., design a counter Need some type of memory to remember the past input values
2	**III**	2	What is Similarity between Combnational & Sequential Circuits Both are a kind of circuit.
3	**III**	3	How Combinational & Sequential Circuits differ from one another? Combinational Circuit is the type of circuit in which output is independent of time and only relies on the input present at that particular instant. On other hand Sequential circuit is the type of circuit where output not only relies on the current input but also depends on the previous output
4	**III**	4	Define Present State and next state. The present state designates the state of flip-flops before the occurrence of a clock pulse. The next state shows the states of flip-flops after the clock pulse.
5	**III**	5	What are the real time applications of Sequential circuits . Used as registers inside microprocessors and controllers to store temporary information. Applied in programmable devices such as CPLD, PLD, and FPGA.
6	**III**	6	What is a Truth Table and excitation table? Truth table:a diagram of the outputs from all possible combinations of input. Excitation table:is a table showing required input conditions for every possible transition of a flip flop.
7	**III**	7	What are asynchronous sequential circuits? Asynchronous sequential circuits do not use clock signals
8	**III**	8	What is the basic one-bit storage device in digital ciruits latch and flip flop
9	**III**	9	How many types of flip flops are there? Name them with their characteristics. there are four types of flip flop 1.SR flip flop-consist of two cross-coupled NOR gates or two crosscoupled NAND gates. It has two inputs labeled S for set and R for reset. 2. D flip flop-D flip flop uses output same as input. 3. JK flip flop-The J input sets the flip-flop to 1, the K input resets it to 0, and when both inputs are enabled, the output is complemented. 4. T flip flop-The T(toggle) flip-flop is a complementing flip-flop and can be obtained from a JK flip-flop when inputs J and K are tied together.
10	**III**	10	What is difference between a latch and a flip flop? latch does not require clock .Flip flop requires clock.
11	**III**	11	What is the characteristic equation of T Flip flop?

			Q(t + 1) = T Q = TQ` + T`Q
12	**III**	12	Which flip flop is known as data flip flop and why? D flip flop is known as data flip flop.Beacuse it will not change the state and store the data present on its output before the clock transition occurred.
13	**III**	13	Enlist the methods that can be used to overcome Race-around condition. If the Clock On or High time is less than the propagation delay of the flip flop then racing can be avoided. This is done by using edge triggering rather than level triggering. If the flip flop is made to toggle over one clock period then racing can be avoided. This introduced the concept of Master Slave JK flip flop.
14	**III**	14	Define Cycle Time. The clock period or cycle time, Tc, is the time between rising edges of a repetitive clock signal.
15	**III**	15	What is hold time? Hold Time: the amount of time the data at the synchronous input (D) must be stable after the active edge of clock.
16	**III**	16	Enlist the types of shift registers.Which one is the fastest? Serial in Serial out (SISO) Shift Register. Serial in parallel out (SIPO) Shift Register. Parallel in Serial out (PISO) Shift Register. Parallel in Parallel out (PIPO) Shift Register. A PIPO register (parallel in, parallel out) is very fast – an output is given within a single clock pulse.

17	**III**	17	How many input lines and output lines are there in a 5 bit Serial in parallel out shift register 5
18	**III**	18	How many clock cycles are required to load and read a 4 -bit data in and out respectively in a Parallel in serial out shift register? The number of clocks needed can be 4 to 5 depending on source data timing and output data timing needs
19	**III**	19	"1101101" is fed in input of a 4-bit Serial-in-parallel-out shift register, what is the output after 6 cock pulses?
20	**III**	20	Mention the number and type of flip flops that will be used in the designing of Mod- 12 counter? 4 flip flop are required and JK flip flop
21	**III**	21	What are the possible range of counts for a Mod- 6 3-bit down counter.
22	**III**	22	Mention the number and type of flip flops that will be used in the designing of Mod- 15 counter? 4 flip flop using jk flip flop

23	**III**	23	What will be the maximum output frequency for a 5-bit ring counter,If the input frquency is 10 MHz 125kHz
24	**III**	24	Mention the number and type of flip flops that will be used in the designing of Mod- 6 counter? 3 flip flop using T flip flop
25	**III**	25	Enlist various types of counters 1.Asynchronous (ripple) counter

			2.Synchronous counter 3.Decade counter 4.Up/down counter 5.Ring counter 6.Johnson counter 7.Cascaded counter
26	**III**	26	What will be the maximum output frequency for a 4-bit jhonson counter,If the input frquency is 20 MHz 3125Hz
27	**III**	27	What is a register? A Register is a collection of flip flops. A flip flop is used to store single bit digital data. For storing a large number of bits, the storage capacity is increased by grouping more than one flip flops.
28	**III**	28	The Term 'loading' a register means The transfer of new information into a register is referred to as loading the register. If all the bits of register are loaded simultaneously with a common clock pulse than the loading is said to be done in parallel.
29	**III**	29	Enlist the steps involved in designing a counter. 1. Description Describe a general sequential circuit in terms of its basic parts and its input and outputs. 2. State Diagram Draw the state diagram for the given sequence. 3. Next State table Develop a next-state table for the specific counter sequence. Using the state diagram as a reference, fill up the present state and next state (yellow) columns. 4. FF transition table 5. K Map Use K-map to derive the boolean expression. 6. Boolean Expression Use the boolean expressions to implement the counter.
30	**III**	30	Whcich flip flop is/are preferably used for designing a counter. JK flip flop

31	**III**	31	What is positive edge triggering in flip flops in positive edge triggered flip flops the clock samples the input line at the positive edge (rising edge or leading edge) of the clock pulse.
32	**III**	32	Why SR flip flop has indeterminate condition? when s=1 and r=1 ,it will give garbage value and this is known as forbidden or inderminate state.
33	**III**	33	What do you understand by term toggling in T-flip flop Toggle operation means if present state is 0 so next state will be 1 and vice versa.
34	**III**	34	Does indeterminate condition for SR flip flop is same for Nand and Nor logics? NO

35	**III**	35	What is the significance of preset and clear inputs in the sequential circuits? The preset input drives the flip-flop to a set state while the clear input drives it to a reset state
36	**III**	36	Predict the output of the J-K fip flop when J=K='1' and clear ='0' Toggle operation will takes place.
37	**III**	37	How many states a ring counter can have for the following preset inputs: a) 0011 b) 0000. Also mention the states. (a) 4 state (b)
38	**III**	38	What will be the output frequency and modulus for an n-bit twisted ring counter?
39	**III**	39	What type of triggering one can use in a sequential circuits? edge triggering.
40	**III**	40	If the clock frequency is 100 MHz, what will be the output frequency for 4bit ring and twisted ring counter respectively?
41	**III**	41	How many states a ring counter can have for the following preset inputs: a) 1111 b) 1000. Also mention the states. (
42	**III**	42	How many input and output lines a 4-bit SIPO and SISO shift register have? of these two which one is faster?
43	**III**	43	How many input and output lines a 4-bit PIPO and PISO shift register have? of these two which one is faster?

44	**III**	44	What counters do? A counter is a device which stores (and sometimes displays) the number of times a particular event or process has occurred, often in relationship to a clock signal. Counters are used in digital electronics for counting purpose, they can count specific event happening in the circuit.
45	**III**	45	Compare sychronous and asyncronous counters on the basis of: a) Principle of operation (b) Design complexity and (c) Delay in signal propagation. 1.Synchronous Counter will operate in any desired count sequence. Asynchronous Counter will operate only in fixed count sequence (UP/DOWN). 2.Synchronous Counter designing as well implementation are complex due to increasing the number of states. Asynchronous Counter designing as well as implementation is very easy. 3.In synchronous counter, propagation delay is less. In asynchronous counter, there is high propagation delay.
46	**III**	46	Give real time applications of Synchronous counters. Alarm Clock, Set AC Timer, Set time in camera to take the picture, flashing light indicator in automobiles, car parking control etc.
47	**III**	47	What are the steps involved in converting one flip flop to another?
			1.Consider the characteristic table of desired flip-flop. 2.Fill the excitation values of given flip-flop for each combination of present state and next state. 3.Get the simplified expressions for each excitation input. If necessary, use Kmaps for simplifying. 4.Draw the circuit diagram of desired flip-flop according to the simplified expressions using given flip-flop and necessary logic gates.
48	**III**	48	Devise the equations for converting S-R Flip flop to D Flip Flop. consider the following characteristic table of D flip-flop. We know that SR flip-flop has two inputs S & R. So, write down the excitation values of SR flip-flop for each combination of present state and next state values. We can use 2 variable K-Maps for getting simplified expressions for these inputs So, we got **S = D & R = D'** after simplifying.
49	**III**	49	What are Finite state Machines The Finite State Machine is an abstract mathematical model of a sequential logic function. It has finite inputs, outputs and number of states.

50	**III**	50	What is moore machine? In the Moore model, the output is a function of the present state only.The Moore model is refereed to as a Moore FSM or Moore machine.
51	**III**	51	Mention features of Mealy Machine. In the Mealy machine, the output is a function of both the present state and input. Less number of states are required. There is more hardware requirement for circuit implementation.
52	**III**	52	Which circuit iout of moore and mealy is more complex to design and why? moore machine is more complex to design because more logic is required to decode the outputs resulting inmore circuit delays.It has more states than mealy machine.
53	**III**	53	What is overlapping

			condition in sequence detectors? In an overlapping sequence detector, the last bit of one sequence becomes the first bit of the next sequence.

54	**III**	54	What is a sequence detector A sequence detector is a sequential state machine that takes an input string of bits and generates an output 1 whenever the target sequence has been detected.
55	**III**	55	How many states ar required to detect the sequence of three consecutive 1's for a mealy design while overlapping is being considered? three distinct
56	**III**	56	What are the steps involved in designing a FSM? 1. Understand the problem Draw a state diagram and statetransition table 2. Determine the machines states Consider missing transitions: Will the machine start? Minimize the state diagram: Reuse states where possible 3. Encode the machines states Encode states, outputs (and inputs, if allowed) as a binary code 4. Design the next-

			state logic Minimize the combinational logic Choices made in steps 2 and 3 can affect the logic complexity 5. Implement the FSM Registered PAL, FPGA, semicustom IC, etc.
57	**III**	57	Why Mealy machines are faster than Moore? Mealy machines react faster to inputs. They generally react in the same clock cycle.. In Moore machines, more logic is required to decode the outputs resulting in more circuit delays. They generally react one clock cycle later.
58	**III**	58	Why Moore machines are more stable? A Moore machine is more stable in this regard, since it only indirectly reacts to input changes. The output only changes when transferring into the next state.
59	**III**	59	How many states are required for detecting sequence of 11011 considering non overlapping for Mealy machine design 5.

60	**III**	60	How many states are required for detecting sequence of 11011 considering non overlapping for Mealy machine design

			5.
61	**III**	61	Give examples of Sequential circuits . Examples of sequential circuits are flip-flop, register, counter, clocks, etc.
62	**III**	62	Define Excitation Table. Excitation table is a table showing reqired input conditions for every possible transitionof a flip flop.
63	**III**	63	What are synchronous circuits a synchronous circuit is a digital circuit in which the changes in the state of memory elements are synchronized by a clock signal.
64	**III**	64	What are synchronous circuits? a synchronous circuit is a digital circuit in which the changes in the state of memory elements are synchronized by a clock signal.

65	**III**	65	Define rise time and fall time Rise time refers to the time it takes for the leading edge of a pulse (voltage or current) to rise from its minimum to its maximum value. ... Conversely, fall time is the measurement of the time it takes for the pulse to move from the highest value to the lowest value.
66	**III**	66	What is Mealy Circuit? A Mealy Machine changes its output on the basis of its present state and current input
67	**III**	67	What is Moore Circuit?

			A Moore Machine's output depends only on the current state. It does not depend on the current input.
68	**III**	68	What is race around condition? For J-K flip-flop, if J=K=1, and if clk=1 for a long period of time, then output Q will toggle as long as CLK remains high which makes the output unstable or uncertain. This is called a race around condition in J-K flip-flop.
69	**III**	69	What do you mean by present state? The present state designates the state of flip-flops before the occurrence of a clock pulse.

70	**III**	70	What do you mean by next state? The next state shows the states of flip-flops after the clock pulse, and the output section lists the value of the output variables during the present state.
71	**III**	71	Which input combination of S & R sets and resets the flip flop? For which input combination there comes indeterminate condidtion? SET when S=1,R=0. RESET when S=0, R=1. INDERMINATE S=1, R=1.
72	**III**	72	What do you mean by latch and FF? The major difference between flip-flop and latch is that the flipflop is an edge-
			triggered type of memory circuit while the latch is a leveltriggered type. Latch does not have clock.Flip Flop have clock.
73	**III**	73	List some of the advantages of synchronous sequential logic circuit. A. Faster operation. B. Case of avoiding problem. C. Lower hardware requirement due to hazards. D. Better noise immunity.

74	**III**	74	List some of the disadvantages of asynchronous sequential logic circuit Some asynchronous circuits may require extra power for certain operations. More difficult to design and subject to problems like sensitivity to the relative arrival times of inputs at gates. ... The number of circuit elements (transistors) maybe double that of synchronous circuits.
75	**III**	75	Why J-K FF used? What are the modification of S-R FF to produce JK FF. The JK flip flop is basically a gated SR flip-flop with the addition of a clock input circuitry that prevents the illegal or invalid output condition that can occur when both inputs S and R are equal to logic level “1”.
76	**III**	76	What is master slave JK FF? The Master-Slave FlipFlop is basically a combination of two JK flip-flops connected
			together in a series configuration. Out of these, one acts as the “master” and the other as a “slave”
77	**III**	77	Why is T -FF called Trigger FF? When both J and K are 1 toggle operation takes place.T flip flop is made out of J-K flip flop.Toggle operation means is presnt state is

			0 so next state is 1 and vice vers.
78	**III**	78	Mention applications of Shift Registers? The shift registers are used for temporary data storage. The shift registers are also used for data transfer and data manipulation. The serial-in serial-out and parallel-in parallelout shift registers are used to produce time delay to digital circuits.
79	**III**	79	What are applications of counters? In time measurement. That means calculating time in timers such as electronic devices like ovens and washing machines. We can design digital triangular wave generator by using counters.
80	**III**	80	What are advantages of asyncronous counters? These are also called as Ripple counters, and are used in low speed circuits. They are used as Divide by- n counters, which divide the input by n, where n is an integer.
81	**III**	81	What are disadvantages of asyncronous counters?

			Counting a large number of bits, propagation delay by successive stages may become undesirably large.
82	**III**	82	Explain operation of Asynchronous Ripple Counter Asynchronous counters are those counters which do not operate on simultaneous clocking. In asynchronous counter, only the first flip-flop is externally clocked using clock pulse while the clock input for the successive flip-flops will be the output from a previous flip-flop.
83	**III**	83	Mention advantages of ripple counter Asynchronous counters can be easily designed by T flip flop or D flip flop. These are also called as Ripple counters, and are used in low speed circuits
84	**III**	84	Mention disdvantages of ripple counter The major disadvantage of ripple counters is that because of new count "rippling" through the flip flops all the bits of the count arrive at different times.

85	**III**	85	What are applications of asynchronous counters? Asynchronous counters are used as frequency dividers, as divide by N counters. These are used for low power applications and low noise emission.
86	**III**	86	What is Jhonson counter? Also mention

			its modulus. A Johnson counter is a modified ring counter, where the inverted output from the last flip flop is connected to the input to the first. It is also called as twisted ring counter. The MOD of the Johnson counter is 2n if n flip-flops are used.
87	**III**	87	Explain the workinng of Jhonson Counter. Johnson counter connects the complement of the output of the last shift register to its input and circulates a stream of ones followed by zeros around the ring. For example, in a 4-register counter, the repeating pattern is: 0000, 1000, 1100, 1110, 1111, 0111, 0011, 0001, so on.
88	**III**	88	What are advantages of jhonson counters? Johnson counter has same number of flip flop but it can count twice the number of states the ring counter can count.

89 90	**III** **III**		90	What are Applications of Ring & Johnson Counters? 89 90	What are disadvantages of jhonson counters? The number of flip flops needed is one half the number of timing signals. What are Applications of Ring & Johnson counters? Parallel to serial data conversion logic circuits. Analog to digital convertors. Digital clocks. Digital triangular wave generator.
					Generating staircase voltage. Frequency divider circuits of the clock signals.
91	**III**			91	In a J-K flip-flop, if J=K, then resulting flipflop, is referred as <u>T flip flop</u>
92	**III**			92	Number of flip flops in modulo-6 Jhonson counter is: 3 flip flops
93	**III**			93	Output frequency for modulo-n Ring counter is:

94	**III**	94	In which kind of sequential circuit master clock is delivered to each module separately? Asynchronous sequential circuit.
95	**III**	95	In which kind of sequential circuit master clock is not delivered to each module separately? Asynchronous sequential circuit.

96	**III**	96	What is difference between synchronous and asynchronous sequential circuits? In synchronous sequential circuits memory elements are used like clocked flip flop. In asynchronous sequential circuits un-clocked memory elements are used like un-clocked flip flop or time delay elements.
97	**III**	97	What is a state machine? A state machine is any device storing the status of something at a given time. The status changes based on inputs, providing the resulting output for the implemented changes
98	**III**	98	How can be state machine represented? state machine can be represented by a directed graph called a state diagram . Each state is represented by a node (circle). Edges (arrows) show the transitions from one state to another. Each arrow is labeled with the input that triggers that transition.
99	**III**	99	What are state equations? A state equation specifies the next state as a function of the present state and inputs.
100	**III**	100	What are state diagrams? The information available in a state table can be represented graphically in the form of a state diagram.In this type of diagram a state is represented by a circle, and the transitions between states are indicated by direscted lines connecting the circles.

101	**III**	101	What are state tables? The time sequence of inputs, outputs, and flip-flop states can be enumerated in a state table (sometimes called transition table).
102	**III**	102	How can we design sequential circuits? Step 1: Create a state transition diagram for the desired sequential circuit. Step 2: Convert the state transition diagram into a state transition table Step 3:Choose flip-flop types & include their exitation tables in the state transition table Step 4:Minimize the functions for the flip-flop inputs Step 5: Use simplified functions for D1 and D2 to design sequential circuit. Step 6: Finally determine the combinatorial circuit to represent the output (if any).
103	**III**	103	What is master reset signal on a flip flop? The master reset (MR) is an asynchronous active LOW input and operates independently of the clock input
104	**III**	104	What difference between the nor based and nand based rs flip flops? The only minor difference occurs because of the properties of a NOR or a NAND gate. The circuit will work in a similar way to the NAND gate

			circuit above, except that the inputs are active HIGH and the invalid condition exists when both its inputs are at logic level “1”.
105	**III**	105	What edge triggered flip flops? Edge triggering is a type of triggering that allows a circuit to become active at the positive edge or the negative edge of the clock signal.
106	**III**	106	What positive edge triggered flip flops? when a flip flop is required to respond at a low to high transition state is known as positive edge triggering
107	**III**	107	What nagative edge triggered flip flops? negative edge triggering-when a flip flop is required to respond at a high to low transition state is known as negative edge triggering
108	**III**	108	What type of triggering is generally preferable for the flip flops which are used for counter design? negative edge triggering
109	**III**	109	Why is negative edge triggered flip flops a better option for counters. The glitches due to race condition can be avoided by using a negative-edge triggered flip-flop instead of the positive-edge-triggered flip-flop used.
110	**III**	110	Which of the flip flop or latch is called a transperant flip flop or latch? D-latch is also known as transparent latch because when the enable inputs is on , the signal propagated directly through the circuit .
111	**III**	111	Why a D flip flop or latch is called a transperant flip flop or latch? D-latch is also known as transparent latch because when the enable inputs is on , the signal propagated directly through the circuit .
112	**III**	112	The flip flops being used as memory elements are useful for which kind of memory? static RAM
113	**III**	113	Can we use flip flops for permanent storage even if power is withdrawn?
114	**III**	114	Give one ic number for a D flip flop. 7475
115	**III**	115	Give one ic number for a RS flip flop. The IC used is SN74HC00N
116	**III**	116	Give one ic number for a JK flip flop. The IC used is MC74HC73A
117	**III**	117	If we have a digital to analog converter and we want a ramp of analog voltage which sequential circuit should be used?

118	**III**	118	What is an up down counter? Counters are used in many different applications. Some count up from zero and provide a change in state of output upon reaching a predetermined value; others count down from a preset value to zero to provide an output state change.
119	**III**	119	Which type of counter should be used for storing the address of the address of the instruction to be executed in the processor? program counter
120	**III**	120	What is difference between latch and flip flop? latch does not have clock. Flip flop have clock.

1	IV	1	What are the two basic types of memory that your computer uses? Ans. Primary memory (RAM and ROM) and Secondary memory (hard drive, CD, etc). Random Access Memory (RAM) is primary-volatile memory and Read Only Memory (ROM) is primary-non-volatile memory.
2	IV	2	EEPROM stands for Ans. electrically erasable programmable read-only memory
3	IV	3	How many 8 k * 1 RAMs are required to achieve a memory with a word capacity of 8k and a word length of eight bits? Ans. Eight
4	IV	4	What is the principal advantage of using address multiplexing with DRAM memory? Ans. It eliminates the requirement for a chip-select input line, thereby reducing the pin count.

5	IV	5	Why is a refresh cycle necessary for a dynamic RAM? Ans. The capacitors tends to discharge with time and must be periodically recharged by refreshing the dynamic memory.
6	IV	6	What is the location of a unit of data in a memory array Ans. Address

7	IV	7	Which memory uses a MOS capacitor as its memory cell? Ans. DRAM
8	IV	8	How many 1K * 4 RAM chips would be required to build a 1K * 8 memory system? Ans. 2
9	IV	9	What is Advantage(s) of an EEPROM over an EPROM Ans. the EEPROM can erase and reprogram individual words without removal from the circuit
10	IV	10	How many address bits are required for a 4096-bit memory organized as a 512 * 8 memory? Ans. 9
11	IV	11	Which type of ROM can be erased by UV light? Ans. The type of ROM which can have its contents erased by ultraviolet light is called EEPROM.

12	IV	12	What is the difference between static RAM and dynamic RAM? | Static RAM	Dynamic RAM
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Made up of flip-flops.	Made up of capacitors.
Large in size.	Small in size.
Data store in the form of voltage.	Data store in the form of charge.
Much expensive as compare to dynamic RAM	Less expensive as compare to static RAM
Low storage capacity	High storage capacity.
Consume more power	Consume less power
Fast	Slow
Data sustain with time.	Data loses with time, so need refreshing circuit*.

13	IV	13	The periodic recharging of DRAM memory cells is called Ans. refreshing
14	IV	14	What is a major disadvantage of the mask ROM? Ans. it cannot be reprogrammed if stored data needs to be changed.
15	IV	15	What is the computer main memory? Ans. Main memory is the primary, internal workspace in the computer, commonly known as RAM (random access memory).

16	IV	16	Which type of ROM has to be custom built by the factory? Ans. All types of ROM are programmable and can be programmed as per requirement but the mask ROM is always programmed for specific application and it can't be reprogrammed.
17	IV	17	How many address lines would be required for a 2K * 4 memory chip? Ans. It means that a memory of 2048 words, where each word is 4 bits. So to address 2048 (or 2K, where K means 2^10 or 1024), you need 11 bits, so 11 address lines
18	IV	18	What does the term "random access" mean in terms of memory? Ans. RAM (Random Access Memory) is the internal memory of the CPU for storing data, program, and program result. It is a read/write memory which stores data until the machine is working. As soon as the machine is switched off, data is erased.
19	IV	19	Which type of ROM can be erased by an electrical signal? Ans. in EEPROM, the ROM can be cleared electrically and thus is less time consuming and more efficient.
20	IV	20	Dynamic memory cells store a data bit in a Ans. Capacitor
21	IV	21	What two functions does a DRAM controller perform? Ans. address multiplexing and the refresh operation

22	IV	22	The technique that extends storage capacities of main memory beyond the actual size of the main memory is called

			Ans. Virtual memory
23	IV	23	A memory in which the information is stored last is on top & is retrieved first is known as Ans. Stack
24	IV	24	What is FPGA Ans. FPGA is a VLSI circuit that can be programmed in the user's location.
25	IV	25	Define Memory unit Ans. The memory unit is a component of a computer system. It is used to store data, instructions and information. It is also known as a principal/primary/internal memory.
26	IV	26	Define Write operation Ans. Memory write operation transfers the address of the desired word to the address lines
27	IV	27	Define Read operation Ans. Memory read operation transfers the desired word to address lines and activates the read control line.

28	IV	28	Give the comparison between PROM and PLA? Ans- PROM: 1.programmable random access memory 2.And gates are fixed and or gates are programmable
			3.Complex architecture 4. Sharing of and gates is possible PLA: 1.programmable logic array 2. OR gates are programmable and AND gates are programmable 3.simpler architecture (depends on the function) 4. Sharing of and gates is not possible
29	IV	29	Why the input variables to a PAL are buffered? Ans. The input variables to a PAL are buffered to prevent loading by the large number of AND gate inputs to which available or its complement can be connected.
30	IV	30	Why was PAL developed? Ans. It is a PLD that was developed to overcome certain disadvantages of PLA, such as longer delays due to additional fusible links that result from using two programmable arrays and more circuit complexity.

31	IV	31	Define PROM Ans. A programmable read-only memory is a form of digital memory where the setting of each bit is locked by a fuse or antifuse.
32	IV	32	Define PLA Ans. A programmable logic array (PLA) is a type of logic device that can be programmed to implement various kinds of combinational logic circuits.

33	IV	33	Define PAL Ans. PAL consist of small programmable read only memory (PROM) and additional output logic used to implement a particular desired logic function with limited components.
34	IV	34	List the major differences between PLA and PAL Ans. PLA: 1.programmable logic array 2. OR gates are programmable and AND gates are programmable 3.simpler architecture (depends on the function) 4. Sharing of and gates is not possible PAL: 1.programmable array logic 2. OR gates are fixed and AND gates are programmable 3. Very simpler architecture if function is written in a optimal way 4. Sharing of and gates is possible

35	IV	35	State the types of ROM Ans. MROM (Masked read-only memory) PROM (Programmable read-only memory) EPROM (Erasable programmable read-only memory) EEPROM (Electrically erasable programmable read-only memory)
36	IV	36	Define address and word Ans.An address is unique identifier assigned to distinguish different devices on a network or the location of data in a computer's memory.
			A word is the natural unit of data used by a particular processor design. A word is a fixed-sized piece of data handled as a unit by the instruction set or the hardware of the processor.
37	IV	37	What is a macrocell? Ans. A macrocell is also generally known as a logic cell. In MAX 3000 and MAX 7000 devices, a macrocell is a basic building block that consists of five product terms and a configurable register, and can be configured individually for either sequential or combinational logic operation.
38	IV	38	Define PLD. Ans. A programmable logic device is an electronic component used to build reconfigurable digital circuits. Unlike integrated circuits which consist of logic gates and have a fixed function, a PLD has an undefined function at the time of manufacture.

39	IV	39	Give the classification of PLDs. Ans- PLDs are classified as PROM (Programmable Read Only Memory), Programmable Logic Array (PLA), Programmable Array Logic (PAL), and Generic Array Logic (GAL)
40	IV	40	What are Flash memories? Ans. Flash memory is an electronic non-volatile computer memory storage medium that can be electrically erased and reprogrammed.
41	IV	41	Give the comparison between PROM and PLA. Ans- PROM: 1.programmable random access memory 2.And gates are fixed and or gates are programmable 3.Complex architecture

			4. Sharing of and gates is possible PLA: 1.programmable logic array 2. OR gates are programmable and AND gates are programmable 3.simpler architecture (depends on the function) 4. Sharing of and gates is not possible

42	IV	42	Give the feature of UV EPROM. Ans. EPROM is individually programmed by an electronic device. After that, the data can be erased by exposing the EPROM to strong ultraviolet light. An EPROM contains a transparent fused quartz window at the top of the package which allows exposure to ultraviolet light.
43	IV	43	Define bit, byte and word. Ans. A bit (short for binary digit) is the smallest unit of data in a computer. A bit has a single binary value, either 0 or 1. A group of binary digits or bits (usually eight) operated on as a unit. a byte considered as a unit of memory size. A word is the natural unit of data used by a particular processor design. A word is a fixed-sized piece of data handled as a unit by the instruction set or the hardware of the processor.
44	IV	44	How many 8 bit words can a 16x8 memory store? Ans-256 words
45	IV	45	Define Capacity of a memory. Ans. The memory capacity is the amount of data a device can store at any given time in its memory.

46	IV	46	Why RAMs are called as Volatile? Ans. RAM (Random Access Memory) is called volatile memory, because in RAM memory gets erased on turning off the power.

47	IV	47	List the two categories of RAMs. Ans. There are two main types of RAM: Dynamic RAM (DRAM) and Static RAM (SRAM).
48	IV	48	Define Cache memory Ans. Cache memory is a small-sized type of volatile computer memory that provides high-speed data access to a processor and stores frequently used computer programs, applications and data.
49	IV	49	What does a dot mean when placed on a PLD circuit diagram? Ans. A point that is programmable.
50	IV	50	Which is the main programming unit in FPGA? Ans. The configurable logic blocks (CLBs) are the basic logic unit of an FPGA.
51	IV	51	Which device is having programmable AND array and fixed OR array Ans- PROM
52	IV	52	In which PLD's both AND and OR array are programmable? Ans- Programmable Logic Array

53	IV	53	What are the functional block in CPLD? Ans- A CPLD contains a bunch of programmable functional blocks (FB) whose inputs and outputs are connected together by a global interconnection matrix
54	IV	54	What does a cross mean when placed on a PLD circuit diagram? Ans- a programmable point
55	IV	55	What is LUT Ans. A Look-Up Table is a discrete block of functionality that can be programmed by the Digital Designer. LUTs use the same truth table concept to relate outputs to inputs.
56	IV	56	Which block in FPGA provides interface between package pins and CLBs Ans- configurable logic blocks
57	IV	57	Give 2 example of SPLD Ans. Simple programmable logic devices (SPLD) are the simplest, smallest and least-expensive forms of programmable logic devices. SPLDs can be used in boards to replace standard logic components (AND, OR, and NOT gates), such as 7400-series TTL.
58	IV	58	What is Sequential Programmable Logic Device. Ans. The SPLD includes flip-flops, in addition to the AND–OR array, within the integrated circuit chip

59	IV	59	What is meant by memory Expansion?

60	IV	60	How many types of PLD is? Ans- Two Types Programmable Logic Array & Programmable Array Logic
61	IV	61	PAL full form is Ans- Programmable array logic
62	IV	62	Name of volatile memory is Ans. RAM (Random Access Memory) and Cache Memory are some common examples of volatile memory.
63	IV	63	The Device having pragrammable AND & fixed OR array is Ans- PROM
64	IV	64	The outputs of PLD is obtained by Ans- OR Gates
65	IV	65	Which logic design is difficult? Ans- Dynamic logic circuits are usually faster than static counterparts, and require less surface area, but are more difficult to design.
66	IV	66	The reprogrammable device is

67	IV	67	USB-type storage device is Ans- USB flash drive is a Secondary storage devices. These typically have broad storage space, and they permanently store data.

68	IV	68	How many address lines required for a 8K memory system? Ans- 8k=2^13. So 2^13 address lines are required.
69	IV	69	Which of the following is the fastest means of memory access for CPU? Ans. Registers are the fastest means of access for CPU. Registers are the small memory locations which are present closest to the CPU.
70	IV	70	Memory that loses its contents when power is lost is Ans- Volatile memory
71	IV	71	What is volatile memory? Ans- Volatile memory is a type of storage whose contents are erased when the system's power is turned off or interrupted. An example of volatile memory is RAM (random access memory).
72	IV	72	What is meant by the term ROM Ans- Read-Only Memory (ROM), is a type of electronic storage that comes built in to a device during manufacturing.

73	IV	73	Which device is used to back up the data? Ans- Floppy Disk, Tape, NEtwork Drive
74	IV	74	Main memory of computer is Ans- Random Access Memory
75	IV	75	What is the permanent memory built into your computer called?

			Ans- ROM
76	IV	76	The chip by which both the operation of read and write is performed Ans. RAM
77	IV	77	Explain PROM. Ans. A programmable read-only memory is a form of digital memory where the setting of each bit is locked by a fuse or antifuse.
78	IV	78	FPGA stands for Ans. field programmable gate array

79	IV	79	Define address and word. Ans.An address is unique identifier assigned to distinguish different devices on a network or the location of data in a computer's memory. A word is the natural unit of data used by a particular processor design. A word is a fixed-sized piece of data handled as a unit by the instruction set or the hardware of the processor.
80	IV	80	Where is FPGA used? Ans. Uses for FPGAs cover a wide range of areas—from equipment for video and imaging, to circuitry for computer, auto, aerospace, and military applications,
81	IV	81	What is mask - programmable? Ans- With a mask programmable PLA, the user must submit a PLA program table to the manufacturer.

82	IV	82	How PLA differs from ROM? Ans. 1. In PLA both AND and OR arrays are configurable unlike in ROM, whereas only the OR gates array is configurable. 2. PLA has the capability to take 'don't care terms' (Boolean overlaps) into account in which ROMs are incapable. 3. ROM has all the combinations of product terms, and therefore, considered as the most general purpose combinational logic device in contrast to PLA, which doesn't have all the combinations.

83	IV	83	Why was PAL developed? Ans- It is a PLD that was developed to overcome certain disadvantages of PLA, such as longer delays due to additional fusible links that result from using two programmable arrays and more circuit complexity.
84	IV	84	Define bit, byte and word Ans- Bit is the abbreviation for binary digit. The BIT is the smallest binary (bivalent) information unit, which can accept a signal status of “1“ or “0“. BYTE- For a unit of 8 binary characters, the term BYTE is used. A byte has the size of 8 bits. A word is a sequence of binary characters, which is regarded as a unit in a specific connection. The word length corresponds to the number from 16 binary characters.
85	IV	85	Define address of a memory? Ans- An address is unique identifier assigned to distinguish different devices on a network or the location of data in a computer’s memory.
86	IV	86	How many words can a 16x8 memory can store? Ans- 256 words

87	IV	87	List the two types of SRAM. Ans- 1. MOFSET 2. Bipolar junction transistor

88	IV	88	Why RAMs are called as Volatile? Ans- Ram is known as volatile memory because it loses all its content when it turns off.
89	IV	89	What is Read and Write operation? Ans- The memory unit supports two basic operations: read and write. The read operation reads previously stored data and the write operation stores a new value in memory. Both of these operations require a memory address.
90	IV	90	Define a bus. Ans- A bus is a subsystem that is used to connect computer components and transfer data between them. For example, an internal bus connects computer internals to the motherboard. A “bus topology” or design can also be used in other ways to describe digital connections. A bus may be parallel or serial. Parallel buses transmit data across multiple wires. Serial buses transmit data in bit-serial format.
91	IV	91	Define PLD. Ans- Programmable Logic Devices PLDs are the integrated circuits. They contain an array of AND gates & another array of OR gates.
92	IV	92	Give the classification of PLDs. Ans- There are three kinds of PLDs: Programmable Read Only Memory Programmable Array Logic

			Programmable Logic Array
93	IV	93	Why the input variables to a PAL are buffered Ans- The input variables to a PAL are buffered to prevent loading by the large number of AND gate inputs to which available or its complement can be connected.
94	IV	94	Explain ROM Ans- ROM stands for Read Only Memory. The memory from which we can only read but cannot write on it. This type of memory is non-volatile. The information is stored permanently in such memories during manufacture. A ROM stores such instructions that are required to start a computer.
95	IV	95	How PLA differs from ROM? Ans- In PLA both AND and OR arrays are configurable unlike in ROM, whereas only the OR gates array is configurable. 2. PLA has the capability to take ‘don’t care terms’ (Boolean overlaps) into account in which ROMs are incapable. 3. ROM has all the combinations of product terms, and therefore, considered as the most general purpose combinational logic device in contrast to PLA, which doesn’t have all the combinations.
96	IV	96	Is PROM reprogramable? Ans- NO. PROMs are used for storage when their content is not expected to change, but in many applications, they have given way to EPROMs and EEPROMs, which can be reprogrammed.

97	IV	97	What is mask - programmable? Ans- With a mask programmable PLA, the user must submit a PLA program table to the manufacturer.

98	IV	98	What is the technique adopted by DRAMs?
99	IV	99	Define PROM. Ans- PROM (Programmable Read Only Memory) It allows users to store data or program. PROMs use the fuses with material like nichrome and polycrystalline. The user can blow these fuses by passing around 20 to 50 mA of current for the period 5 to 20μs.The blowing of fuses is called programming of ROM. The PROMs are one time programmable. Once programmed, the information is stored permanent.
100	IV	100	Why was PAL developed? Ans- It is a PLD that was developed to overcome certain disadvantages of PLA, such as longer delays due to additional fusible links that result from using two programmable arrays and more circuit complexity.
101	IV	101	What does PAL 10L8 specify? Ans- PAL - Programmable Logic Array 10 - Ten inputs L - Active LOW Ouput 8 - Eight Outputs
102	IV	102	What is field programmable logic array? Ans- The second type of PLA is called a field programmable logic array. The user by means of certain recommended procedures can program the EPLA.

103	IV	103	What is PAL? Ans- PAL is Programmable Array Logic. PAL consists of a programmable AND array and a fixed OR array with output logic.
104	IV	104	What is PLA?
			Ans- PLA is Programmable Logic Array (PLA). The PLA is a PLD that consists of a programmable AND array and a programmable OR array.
105	IV	105	Which type of device FPGA are? Ans- FPGA-Field-Programmable Gate Arrays -use read/write memory cells to control the state of each connection.
106	IV	106	What is the main application of FPGAs? Ans- Main FPGA applications are: Medical, video & image processing, telecom & datacom, server & cloud and defense and space. FPGA chips are used in both wired and wireless communications.

107	IV	107	Which type of PLD is more beneficial? Ans- Since PAL consists of programmable AND gates and fixed OR gates and also circuitry working is less.

108	IV	108	Whats is GLA? Ans- Generic Logic Array (GLA) These devices had their properties similar to those of PALs in addition to which they were electrically erasable and re-programmable. This important feature proved to be meritorious as it considerably eased the prototype design which in turn reduced the time to market.
109	IV	109	Is FPGA a PLD? Ans- Yes. FPGA-Field-Programmable Gate Arrays use read/write memory cells to control the state of each connection.
110	IV	110	What is the basic difference between PLA and PAL?

			Ans- The distinction between PLA and PAL is that, PAL have programmable AND array and fixed or array. On the other hand, PLA have programmable AND array and programming OR array.
111	IV	111	How many logic gates are used in PLA? Ans- PLA device architectures are based on the implementation of two logic gate array structures. One array is of Boolean ANDs and the other of Boolean ORs.

112	IV	112	Is FPGA a hardware or software? Ans- an FPGA is a hardware circuit that a user can program to carry out one or more logical operations. Taken a step further, FPGAs are integrated circuits, or ICs, which are sets of circuits on a chip—that's the "array" part. Those circuits, or arrays, are groups of programmable logic gates, memory, or other elements.
113	IV	113	What are Flash memories? Ans- Flash memory is an electronic non-volatile computer memory storage medium that can be electrically erased and reprogrammed.
114	IV	114	List basic types of programmable logic devices. Ans- there are three kinds of PLDs based on the type of arrays, which has programmable feature. Programmable Read Only Memory Programmable Array Logic Programmable Logic Array
115	IV	115	List the three major operations in a flash memory.
116	IV	116	What is a FIFO memory? Ans- First-in, first-out (FIFO) memory chips are used in buffering applications between devices that operate at different speeds or in applications where data must be stored temporarily for further processing.

117	IV	117	Explain the feature of UV EPROM. Ans- EPROM (UV Erasable Programmable Read Only Memory) is a special type of ROM that is programmed electrically and yet is erasable under UV light. The EPROM device is programmed by forcing an electrical charge on a small piece of polysilicon material (called the floating gate) located in the memory cell. When this charge is present on this gate, the cell is "programmed," usually a logic "0," and when this charge is not present, it is a logic "1."
118	IV	118	Define Cache memory. Ans- Cache memory is a small-sized type of volatile computer memory that provides highspeed data access to a processor and stores frequently used computer programs, applications and data.
119	IV	119	Why the input variables to a PAL are buffered Ans- The input variables to a PAL are buffered to prevent loading by the large number of AND gate inputs to which available or its complement can be connected.
120	IV	120	What are the types of ROM.? Ans- Types of Read Only Memory (ROM) – PROM (Programmable read-only memory) – It can be programmed by the user. Once programmed, the data and instructions in it cannot be changed.

			EPROM (Erasable Programmable read only memory) – It can be reprogrammed. To erase data from it, expose it to ultraviolet light. To reprogram it, erase all the previous data. EEPROM (Electrically erasable programmable read only memory) – The data can be erased by applying an electric field, with no need for ultraviolet light. We can erase only portions of the chip. MROM(Marked ROM) – The very first ROMs were hard-wired devices that contained a pre-programmed set of data or instructions. These kind of ROMs are known as masked ROMs, which are inexpensive.

1	**V**	1	What do you mean by logic family?	A range of electronic devices that is made by the same manufacturing technique and provides a number of logic functions	
2	**V**	2	Small Scale integration means?	an integration type for digital circuits that contain transistors numbering in the tens providing a few logic gates per chip	
3	**V**	3	Full form of VLSI?	Very large-scale integration	
4	**V**	4	No of components over a single chip are 10000 or more than the integration is said to be?	VLSI	
5	**V**	5	What is DIP and Full form of DIP?	Dual In-line Package (DIP) is an electronic device package that has electric connecting pins in a rectangular housing and two parallel rows.	
6	**V**	6	What do you mean by bipolar logic family?		
7	**V**	7	What do you mean by MOS logic family?		
8	**V**	8	What do you mean by saturated logic family?		
9	**V**	9	What do you mean by non saturated logic family?		
10	**V**	10	What is the full form of IC?	Integrated circuit (IC	
11	**V**	11	What do you mean by propogation delay?	Propagation delay is the amount of time required for a signal to be received after it has been sent	
12	**V**	12	What do you mean by speed of operation?	Operating speed is sometimes defined as the number of “addition” or “multiplication” operations.	
13	**V**	13	What is power disssipation?	Power dissipation is proportional to the heat generated by the chip or system	
14	**V**	14	What is static power disssipation?	Power consumed when the output or input are not changing or rather when clock is turned off.	
15	**V**	15	What is dynamic power disssipation?	Power consumed during output and input transitions.	
16	**V**	16	What do you mean by fan out?	The fan-out is the number of gate inputs driven by the output of another single logic gate.	
17	**V**	17	What is fan in?	Fan-in is the number of inputs a logic gate can handle	
18	**V**	18	What is noise margin?	Noise margin is a measure of design margins to ensure circuits functioning properly within specified conditions.	
19	**V**	19	How noise margin for a particular circuit is calculated?	The noise margin, NMH = |VOH min – VIH min|	
20	**V**	20	What is the operating temperature range for consumer applications?	70°C for commercial applications, 85°C for industrial applications, and 125°C for military applications.	
21	**V**	21	What is full form of RTL?	Register-transfer level (RTL)	
22	**V**	22	What is full form of RCTL?	Resistor-Capacitor Transistor Logic.	
23	**V**	23	What is full form of DTL?	Diode–transistor logic (DTL)	
24	**V**	24	What is full form of HTL?	High Threshold Logic (HTL)	
25	**V**	25	What is full form of TTL?	Transistor–transistor logic	
26	**V**	26	In how many ways the output from the TTL can be taken?		
27	**V**	27	What is totem pole output configuration?	A type of output structure used with integrated circuits in which one transistor drives the output high while another transistor connected below it pulls the output low	

28	V	28	What is open collector output configuration?	An open collector is a common type of output found on many integrated circuits (IC), which behaves like a switch that is either connected to ground or disconnected.	
29	V	29	What is tri-state output configuration?	Tri-state output (three-state output) An electronic output stage consisting of a logic gate, commonly an inverter or buffer, that exhibits three possible logic states, namely logic 1, logic 0, and an inactive (high-impedance or open-circuit) state.	
30	V	30	The standard gate of TTL is?	NAND	
31	V	31	In which logic family we use multiemitter transistor?	TTL	
32	V	32	Among TTL, CMOS and ECL which is the least power consuming?	CMOS	
33	V	33	Among TTL, CMOS and ECL which is/are the saturated logic family?	TTL	
34	V	34	Among TTL, CMOS and ECL which is the fastest in terms of speed of operation?	ECL	
35	V	35	Among TTL, CMOS and ECL which is/are the non saturated logic family?	ECL	
36	V	36	What is full form of ECL?	Emitter coupled logic	
37	V	37	What is full form of IIL?	Integrated injection logic	
38	V	38	Which IC is used for 2 input NAND gates?	CMOS 4011	
39	V	39	Which IC is used for 2 input NOR gates?	7402	
40	V	40	Which IC is used for 2 input OR gates?	7432	
41	V	41	Which IC is used for 2 input AND gates?	7408	
42	V	42	Which IC is used for inverter?	IC 7404	
43	V	43	IC 7404 used for inverter has how many gates?	Six independent NOT gates	
44	V	44	Which IC is used for 2 input X-OR gates?	7486	
45	V	45	Which IC is used for 2 input X-NOR gates?	74266	
46	V	46	According to moore law the number of transistors will be ..... on a single IC package in 18 months?	Double	
47	V	47	The figure of merit of a logic family is given by the product of:	Propagation delay and power dissipation	
48	V	48	In which family input not connected is equivalent to logic high ?	CMOS	
49	V	49	Among TTL, CMOS, RTL and DTL which logic family as maximum fan out?	CMOS	
50	V	50	External pullup resistor is required in which TTL ouput configuration ?	Open collector ttl	
51	V	51	A TTL circuit acts as a current sink in which state?	Logic – 0 state	
52	V	52	The part nummber of TTL logic IC of basic gates with normal temperature range operation begins with?		

53	**V**	53	Delay which is encountered by signal while transition from input towards output is termed as	Propagation delay
54	**V**	54	The part nummber of TTL logic IC of basic gates with high temperature range operation begins with?	
55	**V**	55	In TTL family if the input is left not connected it is considered as?	unconnected inputs (floating inputs),are left floating then it may become overheated and destroy itself. Thus all the inputs of an IC must be connected to either HIGH or LOW input or OUTPUT of another IC. If floating IC will be left , it will pickup elctromagnetic noise voltages and fluctuate . Thus TTL is connected to logic 1. hence floating input is written as 1.
56	**V**	56	If a logic family has fan-out of 4, then it means?	circuit can drive maximum of 4 inputs
57	**V**	57	MOSFET is also known as	The metal–oxide–silicon transistor (MOS transistor, or MOS)
58	**V**	58	What is the full form of CMOS?	Complementary metal-oxide-semiconductor (
59	**V**	59	Which logic family exhibits its usability for the applications requiring low power consumption?	CMOS - Unipolar family with low power consumption
60	**V**	60	The no of similar logic gates that can be connected to a logic gate output is termed as?	Fan-out
61	**V**	61	In 74LS04, LS stands for	low power schottky
62	**V**	62	74XX series ICs are from which logic family?	Transistor–transistor logic (TTL) logic chips
63	**V**	63	Why are logic families are called so?	A logic family of monolithic digital integrated circuit devices is a group of electronic logic gates constructed using one of several different designs, usually with compatible logic levels and power supply characteristics within a family
64	**V**	64	Name any 4 perfromance specifications of IC logic family?	Speed. Fan-in. Fan-out. Noise Immunity. Power Dissipation
65	**V**	65	If the input is high the TTL logic family NAND gate then the emitter current flows inside the input pin:true or false?	TRUE
66	**V**	66	What are the main three paramters in IC design?	
67	**V**	67	Define Voh?	VOH is the minimum output high voltage of the inverter
68	**V**	68	Define Vih ?	Minimum input voltage that will be recognised as a high input logic level.
69	**V**	69	Define VoL	VOL is the maximum output low voltage of the inverter.
70	**V**	70	Define ViL?	Maximum input voltage that will be recognised as a low input logic level.
71	**V**	71	Define Ioh?	IOH is the current from the pin (sourcing) when the pin has a logic level of one

72	V	72	Define IoL?	IOL is the current into the pin (sinking) when the pin has a logic level of zero
73	V	73	Define Iih?	IIH (high input current) is the leak current when a high level is input to the input pin
74	V	74	Define IiL?	IIL (low input current) is the leak current when a low level is input to the input pin
75	V	75	The standard TTL have following values for the currents: I(OH)= 400uA; I(IH) = 40 uA; I(OL)= 16 mA; I(IL) = 1.6 mA.Fanout will be_____.	
76	V	76	A particular logic family has VOH = 5 V, VOL = 1 V, VIH = 3.5 V and VIL = 2 V. The noise margin values NMH and NML will be	1.5v , 1v
77	V	77	In CMOS logic family which transistor are used as pull up transistor?	PMOS transistors.
78	V	78	In CMOS logic family which transistor are used as pull down transistor?	NMOS transistor
79	V	79	The standard gate of CMOS is?	CMOS NOR Gates
80	V	80	For making a two input NAND gate using NMOS only the two transistors must be connected in?	
81	V	81	For making a two input NOR gate using NMOS only the two transistors must be connected in?	By placing additional out put NMOS transistors in parallel with the output N-Channel MOSFET.
82	V	82	For making a two input NAND gate using PMOS only the two transistors must be connected in?	
83	V	83	For making a two input NOR gate using PMOS only the two transistors must be connected in?	
84	V	84	The propgation delay of CMOS is of the order of ?	The time delay between the 50% crossing of the input and the corresponding 50% crossing of the output.
85	V	85	The propgation delay of TTL is of the order of ?	TTL propagation delay is ~33 ns or less,
86	V	86	The Power dissipation of TTL is of the order of ?	Typically 1.6 mA).
87	V	87	Power dissipation of CMOS is of the order of?	The total power dissipation in a CMOS circuit can be expressed as the sum of three main components: Static power dissipation (due to leakage current when the circuit is idle) Dynamic power dissipation (when the circuit is switching) Short-circuit power dissipation during switching of transistors. 13-Mar-08
88	V	88	If the transistors of any logic family circuit are not driven to the extremes of cutoff and saturation, the family is categorised as?	Non saturated logic family
89	V	89	Which IC is used for 3 input NAND gates?	CD4023
90	V	90	Which IC is used for 3 input AND gates?	74LS11
91	V	91	Which IC is used for 4 input NAND gates?	CD4012 IC
92	V	92	Which IC is used for 4 input AND gates?	74LS21
93	V	93	Which IC is used for 4 input NOR gates?	74HC4002
94	V	94	Which IC is used for 3 input NOR gates?	74LVC1G27

95	V	95	IC used for 2 input NAND gate 7400 has how many gates?	The first part number in the series, the 7400, is a 14-pin IC containing four two-input NAND gates
96	V	96	IC used for 2 input NOR gate 7402 has how many gates?	74LS02 also know as 7402. IT comes up with 4 internal NOR gate. IC 7402 comes up in multiple packages with 14 pins and 2 inputs 4-NOR gates.
97	V	97	IC used for 2 input OR gate 7432 has how many gates?	7432 has 4 OR gates of 2 inputs in 1 package. The internal gates in the ICs are made of Schottky Transistor
98	V	98	IC used for 2 input AND gate 7408 has how many gates?	4 independent gates
99	V	99	What is VCC?	Vcc- voltage common collector is the higher voltage with respect to GND (ground).
100	V	100	What is GND?	GND -ground . it is the reference point of all signals or a common path in electrical circuit where all voltage can be measured from.
101	V	101	In 74xx series ICs the VCC is normally connected to which pin?	14th pin
102	V	102	In 74xx series ICs the GND is normally connected to which pin?	7th pin
103	V	103	To integrate a MOS as a load in CMOS family which two terminals are shorted?	
104	V	104	CMOS ICs which are pin compatible with TTL 74XX series ICs are typically numbered as	
105	V	105	CMOS ICs which are high speed and pin compatible with TTL 74XX series ICs are typically numbered as	
106	V	106	CMOS ICs which are high speed and pin compatible and electrically compatible with TTL 74XX series ICs are typically numbered as	
107	V	107	A transistor that acts as a load for another transistor is called as?	Switch
108	V	108	For a TTL IC if IOH = 200mA and IIH = 10 mA then the FAN OUT is?	
109	V	109	A particular logic family has VOH(min) = 2.4 V, VOL(max) = 0.4 V, VIH(min) = 2.0 V and VIL(max) = 0.8 V. The noise margin will be	
110	V	110	What are the possible logic states in "Tri-State" logic?	Logic 1, logic 0, and an inactive (high-impedance or open-circuit) state.
111	V	111	What is the meaning of high impedance state?	Hi-Z (or High-Z or high impedance) refers to an output signal state in which the signal is not being driven
112	V	112	Which MOS family uses a BJT as well?	The Transistor-Transistor Logic (TTL
113	V	113	BJT is unipolar or bipolar?	Bipolar
114	V	114	MOS is unipolar or bipolar?	Unipolar
115	V	115	Limitation of RTL is?	high power dissipation
116	V	116	Prinicpal disadvantage of RCTL is?	High current dissipation when the transistor conducts to overdrive the output biasing resistor.
117	V	117	In CMOS Power dissipation is at peak during which period?	Dynamic power dissipation
118	V	118	Which logic family uses a zener diode?	DTL(diode transistor logic)

119	**V**	119	Main advantage of CMOS logic family is?	The main advantage of CMOS logic family is their extremely low power consumption	
120	**V**	120	The advantage of using schottky barrier diode in TTL is ?	One of the primary advantages of using a Schottky diode over a regular diode is their low forward voltage drop. This allows a Schottky diode to consume less voltage than a standard diode, using only 0.3-0.4V across its junctions.	